

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME

COVERED CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 40%

Duration: 1 Hour

QUESTIONS: 3

Total Marks:30

Annexure A

Dry Run Evaluation

Code of Conduct:

I _Konkala Reddi Kishore Reddy(192312523)_____
(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Dry Run Evaluation

9. Analyze how Bayes' Theorem helps predict flood-prone areas using rainfall history.

Question 8 – A* Search

A* uses the evaluation function:

$$[$$

$$f(n)=g(n)+h(n)$$

$$]$$

where:

- $(g(n))$ = actual cost from the start node (S) to node (n)
- $(h(n))$ = estimated cost from (n) to goal (G)
- $(f(n))$ = total estimated cost through (n)

Given:

State	$(g(n))$	$(h(n))$	$(f(n)=g+h)$
S	0	7	7
A	2	5	7
B	3	4	7
C	4	3	7
D	5	2	7
E	6	1	7
G	8	0	8

Assuming the displayed graph connects the states in the sequence

[
 $S \rightarrow A \rightarrow B \rightarrow C \rightarrow$
 $D \rightarrow E \rightarrow G$
]

the A* search proceeds as follows.

Iteration Table

Iterati on	OP EN	Expand ed	(g)	(h)	(f)
1	S	—	0	7	7
2	A	S	2	5	7
3	B	A	3	4	7
4	C	B	4	3	7
5	D	C	5	2	7
6	E	D	6	1	7

Iteration	OPEN	Expanded	(g	(h	(f
7	G	E	8	0	8

Final Path

[
 $\boxed{S \rightarrow A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow G}$
]

Final Cost

Since the goal has:

[
 $g(G)=8$
]

the final path cost is:

[
 $\boxed{8}$
]

Note: Your question refers to a "displayed graph," but the graph itself is not included in the message. If that graph has additional branches/edges, the OPEN and expansion order can be different. The above solution assumes the displayed path is (S-A-B-C-D-E-G).

Question 9 – Heuristic Function Analysis

We need to determine which heuristic is admissible.

Definition

A heuristic ($h(n)$) is admissible if it never overestimates the actual cost to reach the goal.

Therefore:

[
 $h(n) \leq \text{Actual Cost}(n)$
]

for every state.

Given:

State	Actual Cost	$(h_1(n))$	$(h_2(n))$
A	10	8	12
B	7	5	6
C	12	10	15
D	5	4	3

Check (h_1)

State A

[
 $8 \leq 10$
]

Admissible.

State B

[
 $5 \leq 7$
]

Admissible.

State C

[
 $10 \leq 12$
]

Admissible.

State D

[
 $4 \leq 5$
]

Admissible.

Therefore:

[
 $\boxed{h_1(n) \text{ is admissible}}$
]

Check (h_2)

State A

[
 $12 > 10$
]

Overestimates the actual cost.

Therefore, (h_2) is not admissible.

State C also confirms this:

[
 $15 > 12$
]

Thus:

[
 $\boxed{h_2(n) \text{ is not admissible}}$
]

Final Answer

Heuristic	Admissible?	Reason
(h_1)	Yes	Never exceeds actual cost
(h_2)	No	Overestimates at A and C

Consequence of Overestimating

If a heuristic overestimates the true remaining cost, A^* can incorrectly consider a path to be more expensive than it really is.

As a result, A^* may:

- Expand the wrong nodes.
- Ignore a potentially optimal path.
- Return a non-optimal solution.

Therefore, with an admissible heuristic, A^* is guaranteed to find an optimal solution under the usual conditions. With an inadmissible heuristic, that optimality guarantee is lost.

Question 10 – A^* vs Greedy Best-First Search

Basic Difference

A^* Search

A^* uses:

$$\boxed{f(n)=g(n)+h(n)}$$

It considers both:

- Cost already travelled ($g(n)$)
- Estimated remaining cost ($h(n)$)

Greedy Best-First Search

Greedy Best-First Search uses only:

$$\boxed{f(n)=h(n)}$$

It chooses the node that appears closest to the goal according to the heuristic.

Comparison

Feature	Greedy Best-First Search	A* Search
Evaluation function	$(f(n)=h(n))$	$(f(n)=g(n)+h(n))$
Considers path cost?	No	Yes
Considers heuristic?	Yes	Yes
Usually faster?	Often	May be slower
Guaranteed optimal?	No	Yes, with admissible heuristic
Main objective	Reach goal quickly	Find lowest-cost path

Example Using the Given Values

For the given states:

State	$(g(n))$	$(h(n))$	A* $(f(n))$
S	0	7	7

State	(g(n))	(h(n))	A* (f(n))
A	2	5	7
B	3	4	7
C	4	3	7
D	5	2	7
E	6	1	7
G	8	0	8

Greedy Best-First Search selects the node with the smallest (h(n)).

Thus, along the assumed graph:

[
 $S \rightarrow A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow G$
]

A* considers (g+h), while Greedy considers only (h).

Comparison Table

Algorithm	Expansion Order	Path	Final Cost
Greedy Best-First		$S \rightarrow A$	8
	$S \rightarrow A \rightarrow B$	$\rightarrow B \rightarrow$	
	$\rightarrow C \rightarrow D \rightarrow$	$C \rightarrow D$	
	$E \rightarrow G^*$	$\rightarrow E \rightarrow$	
A*		G	8
		$S \rightarrow A$	
	$S \rightarrow A \rightarrow B$	$\rightarrow B \rightarrow$	
	$\rightarrow C \rightarrow D \rightarrow$	$C \rightarrow D$	
	$E \rightarrow G^*$	$\rightarrow E \rightarrow$	
		G	

*This order assumes the graph supplied with your question is the simple sequence (S-A-B-C-D-E-G).

Important Point

If the actual displayed graph contains alternative branches, Greedy and A* can produce different expansion orders and

possibly different final costs. Greedy can choose a path with a small heuristic value but a large actual cost, whereas A* considers both ($g(n)$) and ($h(n)$).

Final Summary

[
 $\boxed{\text{Greedy Best-First Search uses only } h(n)}$

]
[
 $\boxed{\text{A* Search uses } g(n)+h(n)}$

] Therefore, A* is generally more reliable for finding the least-cost path, while Greedy Best-First Search may reach a goal faster but does not guarantee an optimal path.